



The Future of Electric Vehicle Charging Infrastructure in India



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India's electric vehicle (EV) revolution is gaining momentum, and at the heart of this transformation lies the crucial development of a robust charging infrastructure. This article delves into the current state and future prospects of EV charging infrastructure in India.

Growth of EV Adoption in India

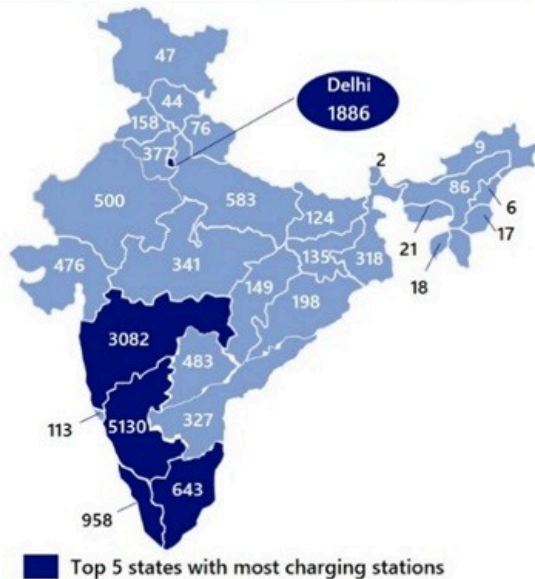
The Government of India aims to achieve 100 percent local production of EVs under its 'Make in India' initiative. However, the Indian EV industry is still in the early stages with a quick growth plan. The EV industry in India is estimated to be worth \$100B by 2030 and it is associated with a mission of net-zero emissions is an initiative associated with driving change or a transformation in the Automobile segment.

State of Public EV Chargers: Location

India has 16,344 charging stations spread across states. Karnataka, Maharashtra, Delhi, Kerala and Tamil Nadu have highest number of charging stations

No of Charging Stations Across Country *

16,344



Bengaluru

4,507



Pune

1,898



Delhi #

1,886



Mumbai#

599



Hyderabad

204



Kochin

199



Trivandrum

187



Chennai

148

Source: *As per BEE, March 2024 ** Map of India shown is only Indicative and may not accurately represent boundaries. UT PCS not indicated on map
Mumbai includes city, suburb & Thane, Bengaluru includes both urban& rural, Delhi limited to state boundary

Tata Power, one of India's largest integrated power companies, announced it had surpassed the installation of **100,000 home EV chargers** in India. **Tata Motors announced** a collaboration with Delta Electronics India and Thunderplus Solutions to establish EV charging stations. In 2024 thus far, Tata has entered deals with oil retailers **Shell** and **HPCL**, which include leveraging the latter company's widespread fuel station network to set up EV charging stations across India. **Hyundai**, **Mahindra & Mahindra**, and **MG Motor** have made similar announcements to install thousands of chargers in the country.



Considering the hitherto weak demand for EVs in India, these initiatives are much needed to allay range anxiety and bolster consumer confidence in EVs. According to [S&P Global Mobility](#), there will be over 9,000 public/semi-public charging points installed in India by the end of 2024. However, India still lacks a robust network of fast chargers. Most public/semipublic chargers are slow chargers. With EVs increasingly equipped with bigger batteries, the charging times in India will only increase if EV users cannot access a fast-charger network.

Current Challenges and Opportunities

- **Infrastructure Expansion:** Despite rapid growth, India's EV charging infrastructure remains inadequate, with only one public charger for every 135 EVs, far below the global average of one charger for every 6 to 20 EVs. The government aims to address this by expanding public and private charging networks across the country.

According to Harish Mehta, CEO of [Jio-bp](#), an EV charging services company, it is very hard for charging point operators (CPOs) to acquire good locations in cities to set up stations where drivers can charge their cars. Jio BP has one of the largest charging networks in India, boasting over 5,000 charging points. Nearly 40% of the network comprises DC fast chargers with more than 60 kW power output.



- **Investment and Innovation:** Recent investments in EV charging start-ups have exceeded \$450 million, highlighting investor confidence in the sector. Innovative models like battery swapping and pay-per-use charging are gaining traction, especially for two- and three-wheelers.

Another challenge is the initial cost of setting up a charging station, which can be prohibitively high in India's Tier 1 cities. Installation and grid upgrade costs can sometimes outstrip hardware costs. According to industry sources, **these peripheral costs can reach INR 500,000 (about USD 6,000) to INR 700,000 (about USD 8,335) for a 60-kW DC charger.**

The high cost of DC chargers and electricity is also cited as an impediment to expanding India's charging network. According to [Servotech Renewable Power System Limited](#), an EV charging solution provider, for chargers with a capacity between 15 kW and 360 kW, the cost can range from INR 300,000 (about USD 3,575) to INR 3 million (about USD 35,700).

Except for a few big players, such as the Tata and Jio groups, the majority of Indian CPOs are startups and small- and medium-sized enterprises. For these companies, charging stations are a long-term play as, with the utilization rate of EV chargers as low as just 5% currently, there is little prospect of an immediate return on the investment.

Emerging Trends in 2025

1. **Ultra-Fast Charging:** The introduction of ultra-fast DC chargers capable of charging EVs in under 30 minutes is expected to reduce range anxiety significantly. Companies like [TATA Power](#),

BluSmart , and **ChargeZone** are leading this push.

2. **Solar and Renewable Energy Integration:** Solar-powered charging stations and wind energy solutions are becoming more prevalent, aligning with India's renewable energy goals.
3. **Battery Swapping and Leasing:** Battery-as-a-Service (BaaS) models are gaining popularity, allowing users to swap batteries quickly rather than waiting for long charging cycles. Companies like **SUN Mobility** and **Bounce Infinity** are leading the way.
4. **Government Initiatives:** The PM E-Drive initiative and state-level incentives are crucial for expanding charging infrastructure, particularly in strategic locations like highways and transport hubs.

Future Projections

- **Installation Targets:** To maintain an optimal ratio of one charger for every 40 EVs, India needs to install over **400,000 charging stations** annually, aiming for a cumulative total of 13.2 lakh by 2030.
- **Market Growth:** The Indian EV market is projected to reach **\$206 billion by 2030**, growing at a CAGR of 49%. This growth will be driven by rising demand for electric two-wheelers and three-wheelers, as well as new four-wheeler models.

India's EV Road To Future

Where It Is Currently Parked

Existing Market Size

\$5.7 Bn

Vehicles Sold

~3 Mn

EV/ICE Ratio

6%

Public Charging Infrastructure

7721 charging stations

Key Players



Key Charging Station Providers



Key Battery Manufacturers



The 2030 Junction Estimated EV Outlook 2030

Estimated Vehicles Sold

10 Mn/year

Estimated Market Potential

\$266 Bn

Public Charging Infrastructure

44K

EV Ratio

30% (4W), 70% for
Commercial Vehicles,
80% (2W)

Source: Vahan, Media Reports

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Conclusion

India's EV charging infrastructure is at the cusp of a new era marked by rapid growth and innovation. With continued government support and industry-led initiatives, the country is poised to overcome the challenges and achieve its ambitious EV adoption goals. The development of a robust and efficient charging infrastructure will be key to ensuring a smooth transition to an electric future in India.

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